

Sanger's early life: From the cradle to the laboratory

Fred Sanger was named after his father, Frederick, who was born in 1876. Educated at St John's College, Cambridge, he qualified as a medical doctor in 1902, and was awarded an MD three years for his post-graduate research at the Pathological Laboratory at Cambridge University under the supervision of the Californian biologist and immunologist George Nuttall. In collaboration with Scotland Yard he developed a forensic test to distinguish between human and animal blood (**Graham-Smith, Sanger 1903**).

On completion of his MD in 1905, Frederick decided to become an Anglican medical missionary in China with the Church Missionary Society (CMS). At this time there were about 3,500 Protestant missionaries in China, most of them in the southern provinces of Hong Kong, Kwangtung and Fukien provinces. Missionary work in China was beset with difficulties. In addition to the linguistic challenge, the indigenous population was highly resistant to Christianity. Many educated Chinese were violently opposed to missionaries and encouraged Chinese authorities to have disputes with them (**Bays, 2011; Church Missionary Society**).

In China, Frederick took up a position as a doctor, like most western trained physicians, in a mission hospital. In addition to his hospital work, he spread the gospel and set up a school for children of the lower classes who, unlike those of the upper mandarin class, had little access to education. Frederick regarded the school as one of his greatest achievements.

In 1912, however, Frederick was forced to leave China due to ill health, possibly tuberculosis. His departure coincided with momentous upheaval in China. Many republican revolutionaries and other groups, including foreign missionaries, rose in protest against the corruption of the ruling elite. The Chinese Qing dynasty was forced

to abdicate in February 1912, marking the end of 2,000 years of imperial rule, and China erupted in a brief civil war(**Gray, 1996**).

On his return to England, Frederick lived for a short while with his widowed mother in Devon before moving with her to Rendcomb, Gloucestershire, in 1913, where he became the local doctor. Three years later, he married Cicely (b.1880), the youngest of six children born to Theodore Crewdson, a wealthy cotton manufacturer who owned mills in Manchester. He had met Cicely when called out to treat her septic finger, a life-threatening condition in the years before antibiotics. By the time the couple met they were both ready to settle down and have a family. Frederick was nearly 40 and Cicely 36.

After their wedding, the couple took up residence in 'The Old House' in Rendcomb, which had room for a small surgery in the 'outhouse' at the side of the house. While very small, the outhouse was sufficient for Frederick's practice because he mostly visited patients in their homes.

Frederick and Cicely had their first child, Theo, in 1917, and in August the following year, Fred was born prematurely. From the start, Fred was much quieter and more introverted than his older brother. Despite their very different temperaments, Fred and Theo were inseparable and remained very close as they grew up. It was Theo's passion for nature that stimulated Fred's interest in science, and he fondly remembered the many hours he and Theo spent hunting for bugs in the family garden and catching newts and tadpoles in its pond. On one occasion Theo caught and caged a grass snake which they found while sitting on a haystack. Theo was also an expert at finding birds' nests. In 1923 the two boys gained a sister, Mary (May)

The three children were brought up to say prayers in the morning and evening and with the religious and moral teachings of Quakerism. Fred's father had become an active Quaker after reading Quaker books inherited by Theodore, his father-in-law, who came

from a long Quaker family line. Unlike his father, however, his mother did not become a Quaker – according to Sanger because she enjoyed singing and listening to hymns.

Sanger spent much of his early childhood exploring the countryside. Until he was five, his family lived in Rendcomb, a picturesque village based in the Cotswolds, Gloucestershire, but moved to Tanworth-in-Arden, a village located in Warwickshire, 12 miles south-east of Birmingham in 1923, in part because Frederick's wish to be nearer to a Quaker community which had a strong presence in Birmingham.

Early education

During his formative years, Sanger was taught by a Quaker governess who came to the family home. Lessons were taught to all three Sanger children, who were joined by various village children. When he was nine, Fred was sent to board at The Downs Preparatory School in Colwall, Herefordshire.

Now known as Downs, Malvern, and part of Malvern College, the school was founded in 1900 on Quaker principles. Its headmaster Geoffrey Hoyland promoted practical learning believing that academic study did not suit all children. In 1924 a miniature railway was added to the school grounds and its steam locomotive thereafter became an important feature of school life (**The Downs Light Railway Trust**).

Initially Sanger found the transition to prep school traumatic. In part this was because this was the first time he had left home, but he also disliked the bullying atmosphere of the school. Sanger, however, found academic work fairly easy so soon rose to near the top of his class. He particularly enjoyed maths but was less keen on Latin.

Bryanston School

In 1932, aged 14, Sanger became a boarder at Bryanston, a new public (i.e. private) boy's school in Dorset. Founded in 1928, the school occupied a palatial country mansion with over 400 acres of land. It was established by J G Jeffrey, a young,

innovative Australian schoolmaster, and was the first English school to adopt the Dalton Plan, a progressive education system first developed in the United States in 1911. This encouraged a flexible timetable. Pupils had some taught lessons together with free periods to carry out independent assignments. They could choose which pieces of academic work they wanted to prioritise, and were expected to keep a daily record of how they used their working and leisure time. Each boy met their tutors weekly and their progress monitored. The relaxed timetable of the school did not suit all the boys, some of whom took advantage of it to do very little work (**Bryanston School, History and Archive**).

Soon after Sanger arrived, the school appointed a new headmaster, Thorold Coade. Sanger welcomed Coade's arrival because he abolished beating and encouraged self-discipline through freedom and self-development. Those who misbehaved were sent on a long run as punishment rather than being beaten. Coade's leadership did not, however, end the very cold baths boys had to take first thing in the morning, a practice continued until 1962 (**Ordish, 2014; Sanger, 2009**).

Sanger described the food at Bryanston as 'grim', but he greatly enjoyed the academic approach of the school. He relished the freedom to work independently and pursue subjects of his choice. It was at Bryanston that he was first introduced to the excitement of science. He had an especially good rapport with Frazer Hoyland, his biology master and the brother of his former headmaster at Downs. Hoyland often took Sanger and other boys on expeditions to carry out biological projects. Sanger also liked spending time in the laboratory making slides.

In addition to Hoyland, Sanger also got on well with his chemistry master, Henry Geoffrey Ordish, who was a softly spoken man with a natural air of gentlemanly authority. Ordish was slightly unconformist, by the standards of the 1930s, in style, thought and manner. He had studied chemistry at Cambridge University and been a researcher in the Cavendish Laboratory. In addition to his scientific skills, Ordish was a considerable mathematician as well as a man of culture who read widely and greatly

enjoyed music and art. Ordish was recruited to Bryanston in its founding year and was its first science master. It had been him who set up the schools' science laboratory. This was located in the old stable block. Ordish electrically wired the laboratory himself.

Ordish, 2014;Sanger, 2009).

Unlike many of his peers Sanger found himself freed from the pressure of exams in his final school year. This was because he took his School Certificate, the school-leaving qualification, a year early, for which he gained seven credits, the highest mark one of Bryanston's pupils had ever achieved. His achievement guaranteed him direct entry into Cambridge. Based on his results he approached his headmaster to see if he could try for a school scholarship. Coade, however, told him in no uncertain terms that he was not up for it. In his usual deprecating manner Sanger later recalled 'I think he was probably right'. Sanger also had the advantage that he did have to worry about scholarship exams for Cambridge because of his family's wealth (Quarrell, 2005).

With a year to do as he wished, Sanger spent many hours in the laboratory experimenting with Ordish with dyestuffs and growing coloured crystals. The laboratory work made a refreshing and highly enjoyable change from sitting and studying books. Working alongside Ordish awakened his desire to pursue a scientific career. What he liked most about experimenting in the laboratory was the chance it gave him to work with his hands, something he frequently did at home, where he was often found absorbed in carpentry and ironwork projects.

Just before Sanger left school he went for half a term on an exchange visit to Schule Schloss Salem, a private boarding school in Southern Germany with close ties to Bryanston. It was established in 1920 along the same liberal principles as Bryanston. Sanger already had some familiarity with German having studied it for his School Certificate, but initially found it difficult to understand and converse with his German peers. He arrived when the Nazis were already well-established, and recalled the ardent fervour of nationalism pervading the school. Three years before Sanger's visit, the

school's headmaster and founder Kurt Hahn, a Jew by birth, had been imprisoned and then forced to flee to England because of his fierce criticism of Hitler and the Nazi regime after his mother witnessed storm troopers killing a young communist. Hahn subsequently founded and became headmaster of Gordonstoun School in Scotland.

Sanger remembered some of the teachers at Salem School dressed in black or brown shirts and the boys doing exercises for the Hitler Youth Movement. While exempt from saluting, Sanger was expected, like all the other boys, to stand up to 'Heil Hitler' before every class. Not being particularly political at the time, Sanger had no inkling of what would follow. His Quaker sensibilities were, however, offended by having to listen to the headmaster, a quiet and friendly man, reading Mein Kampf alongside morning prayers. Nevertheless, he was relatively unaffected by what he had witnessed at the School. As he put it, 'I still had the Quaker philosophy that all men were brothers. People were very nice and ... some of the boys were particularly friendly and weren't all that keen on the Hitler system'.

University

Initially, Sanger considered studying medicine as a career, but he soon abandoned the idea when he saw how hard his father had to work as the local doctor. What he found particularly off-putting was his father's lack of time to focus on any one thing. Science, by contrast, would, he believed, offer him a much quieter and more suitable lifestyle. He thus applied to study science at Cambridge, and was accepted by St John's College, the college both his father and one of his uncles had attended.

In 1936 Sanger began studying chemistry and physics as full subjects and maths and biochemistry as half-subjects for Part I of the Natural Sciences Tripos. He was drawn to biochemistry by the enthusiasm of Ernest Baldwin, one of his tutors at St John's College. What attracted him was Baldwin's suggestion that the exacting scientific methods of chemistry could open up new understandings of biological systems.

At first Sanger struggled with the academic work at Cambridge, taking three years instead of the normal two, to complete Part I of the Tripos. He was partly hampered by the fact that he was way behind his peers in maths and physics, most of whom had done two years' solid work in these areas for the Higher Certificate at School. Sanger had not had this grounding because he had only studied the subjects to General Certificate at school. Having failed to catch up with his contemporaries, Sanger ditched physics in favour of physiology at the end of his first year. Although he decided to continue with maths, he never took to it: his favourite subject was biochemistry, which he studied for Part II of the Tripos in his final fourth year.

Sanger greatly enjoyed the atmosphere in the Department of Biochemistry, which was based in purpose-built accommodation in the Sir William Dunn Institute of Biochemistry. The Department had been founded by Frederick G Hopkins, elected the first Chair of Biochemistry at Cambridge University in 1914. Known by his nickname 'Hoppy', Hopkins's careful experimentation and engaging and friendly personality galvanised everyone in the department. So too did his Nobel Prize, awarded in 1929 in recognition of his discovery of vitamins (**Garcia-Sancho, 2010**).

Sanger also liked the feeling of excitement that existed among Hopkins's young disciples in the department. Many of them were investigating the chemical reactions cells use to process nutrients and other topics, and often discussed their findings during lectures. He particularly enjoyed listening to Malcolm Dixon's lectures on the purification of enzymes and the kinetics of enzyme-catalysed reactions, and those of Joseph Needham who studied morphogenesis, the biological process that determines the shape of an organism. He also found Baldwin's lectures on comparative biochemistry inspiring.

Sanger's religious and philosophical outlook

Sanger did not spend all of his time immersed in academic study, and frequently socialised with the Quaker community in Cambridge. This proved a particularly

important support group for Sanger because, arriving in Cambridge in 1936 in the troubled years before the outbreak of World War II, he often found himself in a minority among his peers because of his strong Quaker pacifist views. Indeed, he had signed up to the 'Peace Pledge Union', a secular pacifist organisation founded in 1934 by Dick Sheppard, canon of St Paul's Cathedral. A large part of the Union's activities involved providing aid for the victims of war in Europe, including those caught up in the Spanish Civil War and Jewish refugees from Germany and Austria. Sanger was also active in the Cambridge Scientists' Anti-War Group, a left-wing pacifist group founded in 1932 to campaign against militarism. Joseph Needham, one of Sanger's biochemistry lecturers, was also a member of the Group.

Over time Sanger gradually lost interest in religion and became an agnostic, but his Quaker upbringing continued to inform his philosophical outlook throughout his life. Nevertheless he continued to be a pacifist in accordance with the Quaker teaching not to take away human life, and the Quaker's injunction to always tell the truth shaped how he behaved. He had such a strong urge to tell the truth that he always found it difficult to lie if feeling unwell when asked how he was. This philosophy also underpinned his attitude to scientific work. He was adamant that scientists should always tell the truth, because as he put it 'a scientist is really studying the truth' (**Sanger, 2001b**).

Marriage

Sanger met his future wife, Joan Margaret Howe, through the Cambridge Scientists' Anti-War Group when he was writing a report about the economic effects of rearmament for the Group. Originally from Leicester, Joan was an economics student at Newnham College. Their courtship began when Sanger was in the middle of studying for his Part II Biochemistry, and Joan provided critical support when Sanger really needed it. Sanger had just lost his parents to cancer, his father in 1937, aged 61, and his mother in 1938, aged 58. Their deaths had been a shock to Sanger, particularly that of his father, who unexpectedly died during a surgical operation.

On December 28 1940 Sanger and Joan married in St Mary's Church located in Syde, a small Cotsworld village where Sanger's favourite uncle, Joseph Dilworth Crewdson, had a farm and house and was the local Justice of the Peace (JP). Sanger had been christened in the church and had spent a lot of his childhood in Syde. The couple went on to have three children, Robin in 1943, Peter in 1946 and Sally Joan in 1960.

War Relief work

Soon after his marriage Sanger completed his undergraduate degree and started working for the war effort. Exempt from military service because he was registered as a conscientious objector, Sanger attended Sipson, a Quaker Relief Training Centre located in Devon. While there he learnt various skills, including building and how to save lives. On leaving Sipson he went to Winford Hospital, a small country hospital near Bristol originally founded to provide orthopaedic treatment to children. By 1940 the Ministry of Health for the Emergency Medical Service had erected seven large hut-type wards on the land surrounding the hospital to accommodate war casualties and extra nursing staff. Serving the hospital as an orderly, Sanger's duties included cleaning floors and toilets. This was his very first paid employment. His salary was just 10 shillings a week, about a quarter of the salary being paid to a private in the British army at this time. He was able to survive on such a small sum because he had free hospital accommodation (**British Military History**).

When Sanger arrived at Winford Hospital the atmosphere was fraught, with wounded soldiers coming in from the evacuation of Dunkirk (May-June 1940) and civilians injured in air raids then frequently taking place over Bristol. Many of the patients he saw had horrific orthopaedic wounds and he was quickly exposed to the realities of war.

Within a short time Sanger realised hospital life was not for him and began looking for ways to return to research. This marked a major shift in his thinking. Pursuing a career in research was very far from his mind when he completed his Cambridge exams. His reticence reflected his lack of confidence, perhaps as a consequence of his struggles

when studying for his Part I exams, for which he had not been awarded any first class marks. His attitude changed when he saw the Cambridge exam results printed in The Times, and learnt, to his astonishment, that he had not only been awarded a first class degree but was only one of the two biochemistry students to have done so.

Doctorate

Inspired by his fine degree results, Sanger wrote to Cambridge to find out whether anyone would take him on as a doctoral student. He had two distinct advantages: he was exempt from military service and had financial support as his wealthy mother had left him a comfortable inheritance. Unable to get any response to his letters, he decided to visit his old department - and soon had a number of potential supervisors vying for his attention. Of these, he found Norman Wingate (Bill) Pirie, a protein chemist, who was attempting to extract edible proteins from grass, the most persuasive and his research the most appealing.

Believing PhD students should be thrown in at the deep end, Pirie immediately handed Sanger a large bucket of frozen mashed-up grass to work with. Sanger had little idea what he should do with the grass which took a very long time to melt. In fact he had not made much progress when a month later Pirie left Cambridge to take up a position at Rothamsted Experimental Station, an agricultural research station in Harpenden.

Following Pirie's departure, Albert Neuberger, a German Jewish refugee, took over Sanger's supervision. Neuberger had fled his country of birth in 1932 just before Hitler became German Chancellor in January 1933. Supported by a grant from the Academic Assistance Council (now the Council for Assisting Refugee Academics), founded in 1933 with funds from British academics, Neuberger did his doctorate in the laboratory of Charles Harrington at University College Hospital in London. Completed in 1936, Neuberger's doctorate demonstrated for the first time that normal proteins contained sugar components. Neuberger joined the Biochemistry Department in Cambridge at the beginning of the Second World War.(Neuberger Sanger, 1942).

For his PhD, Neuberger suggested Sanger study the breakdown products of lysine, an essential amino acid which humans cannot synthesise from scratch. Lysine was of major interest because it could only be sourced from certain food proteins which were frequently absent in the diet of many poor countries. Virtually nothing was known about how the amino acid was broken down and metabolised. To understand this process, Sanger set up a number of experiments feeding lysine to young rats. Frustratingly, however, he could not decipher the metabolism of lysine. It would take another thirty years before scientists could unravel the process.

Despite his lack of conclusive results, Sanger was awarded a doctorate for his research. His thesis, 'The metabolism of the amino acid lysine in the animal body', was examined and passed by Charles Harrington and Albert Charles Chibnall in 1944. In retrospect, Sanger believed that one of the most important skills he acquired from this research was the expertise he gained in amino acid chemistry. This provided an important foundation for his subsequent sequencing work.

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